
ZKM | Hertzlab

MARS!platz

Communication Station

The mission record

MARS — HABITAT ONE

The run	Thursday 15 – Tuesday 27 October 2026 · 13 days · Europe/Berlin
Days recorded	none yet — the run opens Thu 15 Oct 2026
Crew log	0 entries written · 1 states filed
Media sent out	0 files · 0 B · 0 photographs, 0 video, 0 sound, 0 documents
Habitat readings	0 in this record, every one printed · 12 from the station's channels stored in all
Days sealed	0 of 13
Readings log	8 files · 43 kB · every reading ever pulled, in /archive/readings.zip
This record	generated 2026-09-23 11:55 UTC

This record holds only what was entered on the station and what its sensors measured, day by day: the schedule as it was run, the meals, the stores as they were counted, the power and the crew's figures as they were filed, what the crew wrote and sent out, the states filed for them, and every reading of every day — the station's channels, the external node and the hardware, each as it was stored. Nothing in it is generated — no chart, no projection, no total, no figure carried from one day to the next, and no plan for a day that has not come. The messages from Earth and the crew's replies are not part of the record. The photographs are placed in the entries they were sent with; the originals, byte for byte, are in the media ZIP under the hashes printed here.

Contents

The mission	3
The days	5
Today · Wednesday, 23 September 2026 · rehearsal, not the record	6
The crew log	10
Media	11

The mission

MARS — HABITAT ONE. Thursday 15 – Tuesday 27 October 2026, 13 days, on Europe/Berlin time. Three people inside a sealed habitat, and a channel between them and everyone outside it.

Crew

Designation	Role
COMMUNICATION OFFICER	Relay · messages from Earth
SCIENCE OFFICER	Research · habitat systems
HEALTH OFFICER	Crew health · life support

The stores tracked

What the habitat set out with, as written in crew-and-inventory.json. Each day's counts are in that day's record; a store not counted on a day has no figure for it.

Store	Category	Carried in	Unit	Warning below	Critical
Potable water	Consumable	0	L	0	yes
Food rations	Consumable	0	days	0	yes
Oxygen reserve	Life support	0	%	0	yes
CO2 scrubber cartridges	Life support	0	units	0	yes
Battery charge	Power	0	kWh	0	yes
Dust filters	Equipment	0	units	0	
Medical kits	Medical	0	kits	0	
Repair materials	Equipment	0	kg	0	
Germination trays	Research	0	trays	0	

Monitored channels

The habitat's sensor channels as configured in sensors.json: the channel code, what it measures, its unit and the bands the station reads it against.

Channel	Metric	Unit	Expected band	Hard limits	Shown
CH-01	Habitat temperature	°C	18.5 – 24.5	14 – 30	yes
CH-02	Relative humidity	%	32 – 48	20 – 70	yes
CH-03	Carbon dioxide	ppm	400 – 1200	0 – 2500	yes
CH-04	Air pressure	hPa	990 – 1025	960 – 1050	yes
CH-05	Air quality index	AQI	0 – 50	0 – 150	yes
CH-06	Volatile organic compounds	ppb	0 – 400	0 – 1000	yes
CH-07	Particulates PM2.5	µg/m³	0 – 15	0 – 55	yes
CH-08	Radiation dose rate	µSv/h	0 – 0.5	0 – 2	yes
CH-09	Cumulative dose	µSv	0 – 60	0 – 200	yes

How to read this record

Days run in order, and only the days that have happened are here. Each holds the schedule with every task's status as it stands, the meals as entered, the stores counted that day with the figures as they were written, the power and the steps and calories as they were filed, the mission notes, the crew log with its photographs in place, the states filed for the crew with the value chosen and the sentence the station shows for it, what was sent out, the day's sensor summary — each channel's lowest, highest and mean reading and how many readings that is — and then every reading of the day: the station's channels as one row per instant with a column per channel, the external node one row per reading, the hardware one table per device, every value as it was stored, times in habitat time to the second. No figure is totalled, projected or carried from one day to the next, and the messages from Earth and the crew's replies are not part of this record. "The crew log" gathers every blog entry in full, in order; "Media" lists every file with its SHA-256, which can be checked against the ZIP at `/media/export.zip` with `sha256sum`. The same readings, as the station received them — one JSON file per pull, with the stores and figures as they changed and each day's summary — are in the readings log, downloadable whole at `/archive/readings.zip`.

The days

The run opens Thu 15 Oct 2026. No day has been recorded yet; the 13 days are listed below by date. Until then a rehearsal chapter for today follows the list — marked, and not part of the record — so the shape of a day's record can be seen with what there is now.

Day	Date	State	Entries	States filed	Files	Channels	Readings
001	Thursday, 15 October 2026	not yet					
002	Friday, 16 October 2026	not yet					
003	Saturday, 17 October 2026	not yet					
004	Sunday, 18 October 2026	not yet					
005	Monday, 19 October 2026	not yet					
006	Tuesday, 20 October 2026	not yet					
007	Wednesday, 21 October 2026	not yet					
008	Thursday, 22 October 2026	not yet					
009	Friday, 23 October 2026	not yet					
010	Saturday, 24 October 2026	not yet					
011	Sunday, 25 October 2026	not yet					
012	Monday, 26 October 2026	not yet					
013	Tuesday, 27 October 2026	not yet					

Today · Wednesday, 23 September 2026 · rehearsal, not the record

Today, 2026-09-23 · before the run · 3 crew entries · 1 state filed · 0 files sent out · 64 readings

REHEARSAL, NOT THE RECORD. A preview of a day's record with what there is today: today's readings from every source and the states filed today, and whatever has been put into the opening day (SOL 001) so far — its plan, entries, counts, figures and media. This chapter disappears on Thu 15 Oct 2026, when day 001 takes its place.

Schedule

Time	Task	Status
08:00	Shift Change	planned
08:30	Breakfast / Communal Mission Planning	planned
09:30	Science Block 1	planned
11:00	Mental Health / Movement	planned
11:30	Lunch Prep	planned
12:00	Lunch	planned
13:00	Science Block 2	planned
14:00	Science Block 2 / Hot Water Prep with Bike	planned
15:00	Kaffee & Kuchen / Discuss Governance topics	planned
16:00	EVA (Extra Vehicular Activity) / Science Block 3	planned
17:00	EVA (Extra Vehicular Activity) / Science Block 3	planned
17:30	Dinner prep	planned
18:00	Dinner / Washing Up	planned
19:00	Communication Hour / Report Writing Hour	planned
20:00	Social / Personal Time	planned
21:00	Social / Personal Time	planned
22:00	Shift Change / Lights Out	planned

Meals

Slot	Meal	kcal	Water L	Prep min	Wh
Breakfast	Rehydrated oats, dried apple Oat base 80 g · Dried apple 20 g · Water 0.35 L	420	0.35	8	90
Lunch	Lentil and tomato stew Lentil block 120 g · Tomato powder 25 g · Water 0.60 L	610	0.6	22	260
Dinner	Rice, beans, preserved greens Rice 110 g · Bean mix 90 g · Preserved greens 40 g	720	0.75	28	310

Stores counted that day

No store was counted on this day.

Crew log

COMMUNICATION OFFICER

Written 2026-09-23 13:55 habitat time
Today was kind of shit.

SCIENCE OFFICER

Written 2026-09-23 13:55 habitat time
no Science happened.

Science was sciencing
wsdfedfgdrgvwGFRRGVGV

tHIS IS WHATEVER THIS IS.

HEALTH OFFICER

Written 2026-09-23 13:55 habitat time
Everyone Had kind of shit healtj.

Crew states filed

The value is the face chosen in mission control, 0 calm to 100 angry; the sentence is what the station shows for it.

Time	Officer	Value	Condition	Shown as	Activity	Filed by
13:55	SCIENCE OFFICER	40	LEVEL	level — neither calm nor cross	Rehearsal	control

Habitat

Each channel's lowest, highest and mean reading over the day, and how many readings that is. Every reading itself is in the readings log.

Channel	Metric	Low	High	Mean	Unit	Readings	Sealed
CH-01	Habitat temperature	21.4	21.9	21.7	°C	6	open
CH-03	Carbon dioxide	640	685	662.5	ppm	6	open

External sensor node

Channel	Low	High	Mean	Unit	Readings
CO2 · external node	600	678	639	ppm	40
Temperature · external node	20.50	20.80	20.65	°C	40
Humidity · external node	41	41	41	%	40
Pressure · external node	1013.20	1013.20	1013.20	hPa	40
Light · external node	90	129	109.50	raw	40
Node battery	3.90	3.90	3.90	V	40

Habitat hardware (Home Assistant)

Device	Low	High	Mean	Added today	Unit	Readings
Cricket Terrarium Temperature	26.8	27.1	27.0	—	°C	12

Every reading of the day

64 readings, each as it was stored. Times are habitat time, to the second.

The station's channels · 12 readings

CH-01 °C = Habitat temperature · CH-03 ppm = Carbon dioxide

Time	CH-01 °C	CH-03 ppm
08:55:25	21.4	640
09:55:25	21.5	649
10:55:25	21.6	658
11:55:25	21.7	667
12:55:25	21.8	676
13:55:25	21.9	685

The external sensor node · 40 readings

Time	CO2 ppm	Temperature °C	Humidity %	Pressure hPa	Light raw	Battery V	Signal dBm
00:00:00	600	20.5	41	1013.2	90	3.9	-60
00:20:00	602	20.6	41	1013.2	91	3.9	-60
00:40:00	604	20.7	41	1013.2	92	3.9	-60
01:00:00	606	20.8	41	1013.2	93	3.9	-60
01:20:00	608	20.5	41	1013.2	94	3.9	-60
01:40:00	610	20.6	41	1013.2	95	3.9	-60
02:00:00	612	20.7	41	1013.2	96	3.9	-60
02:20:00	614	20.8	41	1013.2	97	3.9	-60
02:40:00	616	20.5	41	1013.2	98	3.9	-60
03:00:00	618	20.6	41	1013.2	99	3.9	-60
03:20:00	620	20.7	41	1013.2	100	3.9	-60
03:40:00	622	20.8	41	1013.2	101	3.9	-60
04:00:00	624	20.5	41	1013.2	102	3.9	-60
04:20:00	626	20.6	41	1013.2	103	3.9	-60
04:40:00	628	20.7	41	1013.2	104	3.9	-60
05:00:00	630	20.8	41	1013.2	105	3.9	-60
05:20:00	632	20.5	41	1013.2	106	3.9	-60
05:40:00	634	20.6	41	1013.2	107	3.9	-60
06:00:00	636	20.7	41	1013.2	108	3.9	-60
06:20:00	638	20.8	41	1013.2	109	3.9	-60
06:40:00	640	20.5	41	1013.2	110	3.9	-60
07:00:00	642	20.6	41	1013.2	111	3.9	-60
07:20:00	644	20.7	41	1013.2	112	3.9	-60
07:40:00	646	20.8	41	1013.2	113	3.9	-60
08:00:00	648	20.5	41	1013.2	114	3.9	-60
08:20:00	650	20.6	41	1013.2	115	3.9	-60
08:40:00	652	20.7	41	1013.2	116	3.9	-60
09:00:00	654	20.8	41	1013.2	117	3.9	-60
09:20:00	656	20.5	41	1013.2	118	3.9	-60
09:40:00	658	20.6	41	1013.2	119	3.9	-60
10:00:00	660	20.7	41	1013.2	120	3.9	-60
10:20:00	662	20.8	41	1013.2	121	3.9	-60
10:40:00	664	20.5	41	1013.2	122	3.9	-60
11:00:00	666	20.6	41	1013.2	123	3.9	-60
11:20:00	668	20.7	41	1013.2	124	3.9	-60
11:40:00	670	20.8	41	1013.2	125	3.9	-60
12:00:00	672	20.5	41	1013.2	126	3.9	-60
12:20:00	674	20.6	41	1013.2	127	3.9	-60
12:40:00	676	20.7	41	1013.2	128	3.9	-60
13:00:00	678	20.8	41	1013.2	129	3.9	-60

Cricket Terrarium Temperature · sensor.sonoff_temperatur_temperature_1 · 12 readings

Time	Value (°C)	As reported
00:00:15	26.8	26.8
01:00:15	27.1	27.1
02:00:15	26.8	26.8
03:00:15	27.1	27.1
04:00:15	26.8	26.8
05:00:15	27.1	27.1
06:00:15	26.8	26.8
07:00:15	27.1	27.1
08:00:15	26.8	26.8
09:00:15	27.1	27.1
10:00:15	26.8	26.8
11:00:15	27.1	27.1

The crew log

Everything the crew wrote, in full, day by day and officer by officer — the same text as in each day's record, gathered so the log can be read straight through. Photographs sit where they were placed in the entry; video and sound are listed. An entry held back from the public station is included and marked, because this is mission control's record.

Nothing has been written yet.

Media

Every file the crew sent out: 0 visible (0 B). Each is stored once under its SHA-256 in the station-data volume and is in /media/export.zip under its filename; sha256sum on any copy must give the hash printed here.

Nothing has been sent out.